

Two-Button LED Pattern Controller

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What this is

This Arduino sketch lets you control a strip or group of LEDs using two simple push buttons.

- Button A: go FORWARD to the next pattern.
- Button B: go BACKWARD to the previous pattern.
- The patterns are pre-written for you.
- You only need to change the numbers at the TOP of the file (the configuration section).

Who this is for

This is for people who are comfortable with electronics (wiring, power, LEDs) but are not super comfortable editing code. Think: "I know what pin 6 is, but don't make me read 200 lines of C++."

How it works (high level)

1. Arduino starts and sets up the LED pins and the two button pins.
2. It stores all the patterns in a list (array).
3. Every time you press Button A, the sketch moves to the NEXT pattern.
4. Every time you press Button B, the sketch moves to the PREVIOUS pattern.
5. The 'loop()' function runs the CURRENT pattern over and over, until you change it again.
6. Some patterns use a delay value (speed). You can change that at the top.

What you are allowed to change

Only edit the CONFIGURATION SECTION at the top of the .ino file.

Typical things to change there:

- LED_PIN : the Arduino pin that goes to your LEDs
- NUM_LEDS : how many LEDs you really have
- BUTTON_NEXT_PIN : which pin your "next" button is on
- BUTTON_PREV_PIN : which pin your "previous" button is on
- PATTERN_DELAY_MS : how fast the animation runs
- BRIGHTNESS : if using an addressable LED library

Do NOT change the logic in loop(), do NOT change the pattern-switching code, and do NOT change the button-reading helper functions, unless you know exactly what you are doing.

How to add another pattern

1. Scroll down to the PATTERN FUNCTIONS section.
2. Copy an existing pattern function (for example: runChasePattern()).
3. Paste it right below.
4. Rename it to something meaningful, like runRedPulse().
5. Inside the new function, change only the LED-writing parts and the delay.
6. Go to the array/list of patterns and add your new function name there.

7. Upload again.

Common customizations

- "It's too fast." --> Increase PATTERN_DELAY_MS (for example from 75 to 110).
- "It's too slow." --> Decrease PATTERN_DELAY_MS (for example from 75 to 35).
- "Swap buttons." --> Just swap the pin numbers for BUTTON_NEXT_PIN and BUTTON_PREV_PIN.
- "More LEDs." --> Change NUM_LEDS.
- "Different Arduino pin." --> Change LED_PIN.

Button wiring (very short)

- Each button should go from the pin to GND, and use INPUT_PULLUP in code.
- That means "pressed" will read as LOW.
- Don't worry: the code already handles this.

Troubleshooting

- Nothing lights up: check LED_PIN and power.
- Button does nothing: check the button pins and that you are using INPUT_PULLUP wiring.
- Only one pattern works: make sure the patterns array includes ALL pattern functions.

License / usage

Use it, change it, share it with your lab group. Just keep the top comments so everyone knows how to customize it.